When Annotators Disagree: A Controlled Evaluation of Gender Bias in Sentiment Analysis Using Synthetic Datasets

Abstract. *This study investigates gender-related annotation bias in sentiment classification. First, we introduce a controlled synthetic dataset generation method that simulates parallel male and female sentiment labels with adjustable inter-annotator agreement. Then, we present the Gender Comparison Methodology, which trains classifiers separately on gender-partitioned labels and evaluates their predictions using shared textual inputs. Agreement is assessed using metrics such as Cohen's Kappa, chi-square tests, and Cramér's V. Results show that even moderate disagreement between annotators leads to systematic model divergence, highlighting the importance of annotator identity in shaping classification behavior and informing fairness-aware auditing practices.*

1. Introduction

Language is both a collective and individual construct. While shared linguistic systems enable communication (de Saussure 1959), individual interpretations are shaped by personal experiences and social identity. In Sentiment Analysis (SA), such variation is often ignored in favor of consensus-based datasets, despite growing evidence that annotator characteristics—including gender—can influence labeling decisions (Kenyon-Dean et al. 2018; Biester et al. 2022; Jiang et al. 2024). Hence, gender-based differences in perception and communication shape how individuals interpret emotions and express sentiments. In the context of SA, such divergences can influence both, the data annotation and the predictions of machine learning systems, that might carry biases from individuals social perspectives.

Meanwhile, the hype surrounding Generative AI often obscures these challenges, portraying Large Language Models (LLMs) as silver-bullet solutions. Yet, models trained on crowd-annotated data can reproduce and amplify societal bias (Kiritchenko and Mohammad 2018; Levshina et al. 2024), especially when disagreement is collapsed into single "ground truth" labels.

Sentiment classification models often rely on datasets annotated under the assumption of objective, homogeneous labeling. However, annotation is not a purely mechanical task—involving deep interpretation shaped by sociocultural factors. Hence, controlling and analyzing such divergences is not only a technical challenge but also an ethical imperative since annotations may carry implicit biases reflecting the annotators' identities, including gender, age, or sociopolitical attitudes (Kenyon-Dean et al. 2018; Biester et al. 2022; Jiang et al. 2024).

Nevertheless, it is also important to notice that many studies in the literature do not report the gender composition of annotators, leaving unanswered whether divergences in labels may have demographic roots. For instance, widely cited datasets often refer to annotators generically as "crowdworkers" without disclosing gender distribution (Kenyon-Dean et al. 2018; Kiritchenko and Mohammad 2018; Alves et al. 2022). This omission makes it more difficult to assess how much prejudice might result from gendered interpretation. This brings us to a critical hypothesis: what if two sentiment classifiers

were trained independently, one using labels solely from male annotators and the other exclusively from female annotators? Which leads us to the subsequent research inquiries: (RI1) Would there be a systematic difference in their predictions?; (RI2) What would this difference tell us about the social aspects of language interpretation?

To address this challenge in a controlled and replicable way, we propose a synthetic dataset generation methodology that enables fine-grained control over annotator agreement and label diversity. Building on this, we introduce a gender comparison methodology that simulates male and female annotations independently to assess how classification models respond to divergent labeling sources. This dual contribution aligns with perspectivist principles (Plank 2022) and advances fairness-oriented evaluation in SA.

This article is organized as follows. Section 2 explains the theoretical background. Section 3 reviews prior research on gender bias in natural language processing. Section 4 details the synthetic dataset generation methodology. Section 5 introduces the gender comparison methodology. Section 6 describes the experimental setup and statistical interpretation guidelines. Section 7 presents the main results and their analysis. Finally, Section 8 concludes the study with a summary of the findings, a discussion of limitations, and directions for future work.

2. Theoretical Background

As a subfield of Natural Language Processing (NLP), SA is fundamentally shaped by human perception and directly influenced by social identity, ideology, and lived experience (Biester et al. 2022). It is not purely computational, but inherits social aspects of the human annotator. And, although the task of assigning sentiment text polarities (positive, negative, or neutral impressions), may appear objective, in reality the annotation decisions are often influenced by implicit biases and cultural perspectives.

Annotators' ideological attitudes, such as neosexist or authoritarian views, can influence how they label text (Jiang et al. 2024). These biases are often embedded in benchmark datasets, especially when majority voting masks annotation disagreement (Kenyon-Dean et al. 2018). As a result, classifiers trained on such data may amplify existing inequalities rather than mitigate them.

Gender, in particular, has been consistently identified as a key source of variation in annotation, influencing not only the perception of emotional tone but also judgments of offensiveness, sarcasm, and intent (Kiritchenko and Mohammad 2018; Jiang et al. 2024). Recent research show such biases across various NLP benchmarks, including tasks like abusive language detection, toxic comment classification, and sexism recognition.

Annotators' ideological attitudes and tendencies towards right-wing authoritarianism or neosexist beliefs affect the way they label content, leading to divergent interpretations of what constitutes sexist or harmful speech (Jiang et al. 2024). These discrepancies are not just data noise, instead they represent valid, but also conflicting, perspectives frequently lost in the traditional dataset construction processes (Jiang et al. 2024).

This problem aggravates in large-scale annotation pipelines where labels are aggregated via majority voting, effectively silencing minority points of view, and giving to the complex and suitable aspects of sentiments simple monolithic "ground truth" values. As shown by (Kenyon-Dean et al. 2018), circa 30% of sentiment annotations in crowd-

sourced datasets can be classified as "controversial" or subject to annotator disagreement. Yet these instances are often discarded or force-labeled to maintain dataset consistency.

The result is a modeling landscape where even highly accurate systems replicate and amplify the biases embedded in the training data. These models may perform adequately on average but tend to under-perform when applied to content produced in marginalized communities (Kiritchenko and Mohammad 2018; Levshina et al. 2024).

Hence, with this scenario, rethinking the way data is created, annotated, and interpreted is important for better fairness in machine learning and artificial intelligence model development. Unlike traditional datasets, which are constrained by human annotator availability and budget limitations, synthetic datasets can be programmatically created to incorporate controlled levels of label variation and demographic diversity.

Recent advances such as SentiGEN (Sundarreson and Kumarapathirage 2024) propose synthetic dataset methodologies to overcome these limitations. By enabling the generation of balanced datasets with controlled agreement levels, they allow researchers to simulate diverse labeling conditions—particularly valuable in subjective tasks like sentiment classification, where interpretation is shaped by cultural and gendered perspectives. SentiGEN leverages fine-tuned transformers and evolutionary optimization, showing that models trained on synthetic data can rival or even surpass those trained on real-world data, especially in capturing sentiment nuance and mitigating majority-class overfitting.

Hence, the need for perspectivist approaches is underscored by mounting evidence that demographic attributes—such as gender, race, and language—can significantly shape both human annotation and model behavior. Studies have shown that SA systems may assign different emotional intensities to otherwise identical sentences depending solely on the gender or race of the referent (Kiritchenko and Mohammad 2018; Jiang et al. 2024). In multilingual contexts, these asymmetries become even more complex, as cross-linguistic variation introduces additional layers of interpretation and bias (Levshina et al. 2024). Recent findings further suggest that generative AI systems, especially those based on large language models (LLMs), not only reproduce but may amplify such biases, embedding them into seemingly neutral outputs. These results highlight the urgency of evaluating model fairness through frameworks that account for the subjective and socially situated nature of meaning.

Recent research has highlighted an emerging paradigm in the NLP community that advocates for perspectivist data practices (Biester et al. 2022; Plank 2022). These approaches challenge the traditional notion of a single correct label, instead promoting the preservation of multiple, coexisting annotations to reflect the inherently subjective nature of interpretation. Such a perspective is particularly relevant to SA, where emotion, tone, and social context intertwine in complex and culturally mediated ways (Biester et al. 2022; Plank 2022). Additionally, (Assi and Caseli 2024) demonstrate that even advanced generative models like GPT-3.5 Turbo exhibit systematic divergences in output when prompted with gender-marked phrases in both Portuguese and English, reinforcing the persistence of implicit bias across linguistic and cultural contexts.

3. Related Work

Past research has consistently demonstrated the systematic influence of gender on SA. The Equity Evaluation Corpus (EEC) (Kiritchenko and Mohammad 2018) revealed per-

sistent discrepancies in model predictions based on gendered terms. (Jiang et al. 2024) further showed that annotators' ideological positions significantly shape labeling outcomes, particularly in tasks involving gender-based violence. Similarly, user demographics such as gender and age have been found to affect both the expression and interpretation of sentiment in product reviews (Kumar et al. 2020).

Synthetic data generation has gained traction as a tool for fairness-oriented experimentation. Resources like SentiGEN (Sundarreson and Kumarapathirage 2024) allow researchers to simulate and control annotation bias across multiple conditions. In parallel, perspectivist NLP frameworks advocate for the retention of multiple interpretations rather than enforcing a single canonical label (Biester et al. 2022; Plank 2022). In this spirit, disagreement is revalued as a source of insight rather than noise. (Biester et al. 2022) and (Kenyon-Dean et al. 2018) caution against discarding "controversial" examples, which may encode valuable variance that contributes to model robustness.

Levshina et al. (Levshina et al. 2024) also challenge the assumption of uniform sentiment polarity across gendered terms, revealing ambivalence rather than pejoration in various linguistic contexts. Likewise, ideological and demographic attributes of annotators have been shown to significantly influence labeling, reinforcing the case for demographic-aware annotation pipelines (Jiang et al. 2024).

Despite these insights, few studies have explicitly simulated gendered annotations for systematic experimental comparison. The present work addresses this gap by introducing a controlled synthetic framework designed to assess how gender-specific labeling affects classifier behavior and performance.

4. Synthetic dataset methodology

The synthetic dataset generation procedure is structured around three modular functions, each responsible for a distinct stage of construction. Algorithm 1 outlines this process.

Algorithm 1: Synthetic Sentiment Dataset Generation

Input: N `// Total number of instances (multiple of 3)`
Input: $\gamma \in [0, 1]$ `// Desired concordance rate`
Input: $L_{\min}, L_{\max}$ `// Minimum and maximum text lengths`
Output: Dataset $\mathcal{D} = \{(t_i, m_i, f_i)\}_{i=1}^{N}$

$\mathcal{C} \leftarrow \{\texttt{negative}, \texttt{neutral}, \texttt{positive}\}$; `// Sentiment classes`
$n_c \leftarrow N/3$; `// Instances per class`
$M, F \leftarrow \texttt{GenerateBalancedLabelPairs}(\mathcal{C}, n_c, \gamma)$; `// Balanced and partially concordant`
$T \leftarrow \texttt{GenerateSyntheticTexts}(N, L_{\min}, L_{\max})$; `// Synthetic textual inputs`
$\mathcal{D} \leftarrow \texttt{AssembleDataset}(T, M, F)$; `// Triplets: text, male label, female label`
$\texttt{Shuffle}(\mathcal{D})$; `// Randomize order`
return $\mathcal{D}$

The process illustrates how balanced label pairs are generated with controlled concordance, how synthetic text samples are produced with variable lengths, and how the final dataset is assembled by combining these elements into triplets of text and gendered

labels. The algorithm receives three key inputs: the total number of instances N (which must be a multiple of 3 to ensure class balance), a target inter-annotator concordance rate $\gamma \in [0, 1]$, and the parameters $L_{\min}$ and $L_{\max}$, which define the minimum and maximum token length of the generated texts, respectively.

The first function, $\texttt{GenerateBalancedLabelPairs}(\mathcal{C}, n_c, \gamma)$, creates two parallel sequences of sentiment labels—denoted $M = (m_1, \ldots, m_N)$ for male and $F = (f_1, \ldots, f_N)$ for female annotators—based on a fixed class set $\mathcal{C} = \texttt{negative}, \texttt{neutral}, \texttt{positive}$. Each class is represented in equal proportion ($n_c = N/3$ per class), and concordant label pairs are assigned according to the proportion defined by γ. For the remaining discordant instances, label combinations are generated in a way that preserves the overall class balance in both M and F, while ensuring that the two sequences diverge in annotation.

The second function, $\texttt{GenerateSyntheticTexts}(N, L_{\min}, L_{\max})$, constructs a list $T = (t_1, \ldots, t_N)$ of synthetic text samples. Each t_i is formed by randomly sampling between $L_{\min}$ and $L_{\max}$ placeholder tokens (e.g., `word1`, `word2`, ...). Although devoid of semantic content, these sequences emulate realistic variation in text length and provide a neutral input substrate for isolating the effects of label-based differences on model behavior.

Finally, the `AssembleDataset` function produces the final dataset $\mathcal{D} = (t_i, m_i, f_i)_{i=1}^{N}$ by associating each synthetic text t_i with its corresponding male label m_i and female label f_i. The triplets are then shuffled uniformly to remove any ordering bias, resulting in a ready-to-use dataset suitable for downstream training and evaluation tasks. This procedure offers precise control over annotator agreement, text variability, and label assignment, providing a principled experimental framework for analyzing the interaction between annotation diversity and classifier fairness in SA.

5. Gender comparison methodology

The methodology employed in this study (Fig. 1) aims to systematically assess the divergences in model behavior resulting from gender-specific annotations in SA datasets. The core premise involves evaluating how classifiers trained on identical textual inputs but labeled differently—according to male or female annotators—produce predictions that may reflect or amplify underlying biases.

The process begins with a SA dataset in which each text instance is associated with two independent labels: one assigned by male annotators and the other by female annotators. These labels are categorical (e.g., negative, neutral, or positive).

In Step 1, feature extraction is applied to transform the raw text into numerical representations suitable for machine learning models. Various vectorization techniques may be employed, including term frequency (TF), term frequency-inverse document frequency (TF-IDF), or other bag-of-words-based methods, depending on the experimental setup. Importantly, the feature space remains shared across all subsequent models to ensure comparability.

Step 2 involves the independent training of two distinct classifiers: one using the male-labeled classes as ground truth and the other using the female-labeled classes. Both models are trained on the same set of input texts and feature representations, differing only

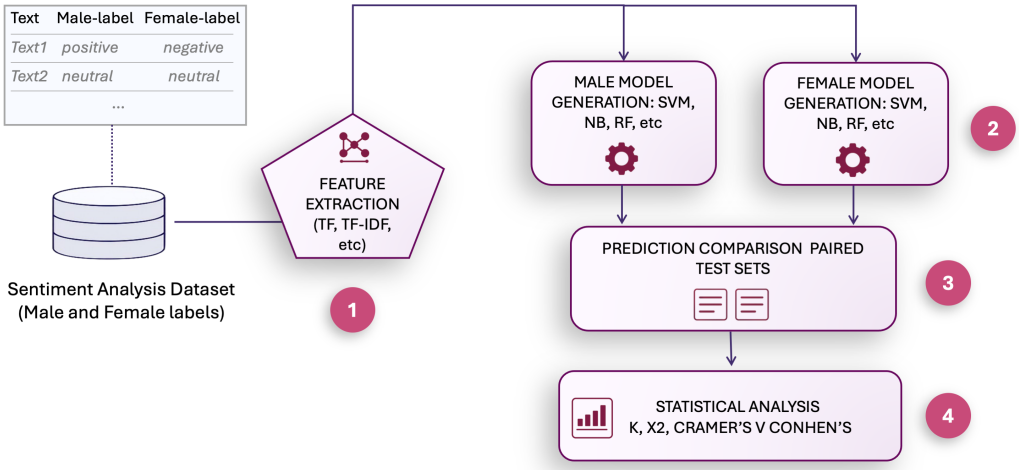

Figura 1. Gender comparison methodology

in the annotation source used for supervision. This design ensures that any divergence in the models' learned decision boundaries can be attributed solely to the difference in gendered annotations, rather than variations in the input data or architecture.

In Step 3, classifiers are evaluated on paired test sets—identical in content but labeled according to gender. Predictions from the male-based model and the female-based model are compared using the Cohen's Kappa coefficient, which quantifies the agreement between the two classifiers beyond chance. This comparison is conducted across multiple folds in a cross-validation setting to ensure robustness and mitigate sampling variance.

Finally, in Step 4, a set of statistical analyses is conducted to quantify and interpret the divergences between male-based and female-based models. These include the proportion of agreement between annotators, which provides an intuitive measure of consensus, and Cramér's V, which assesses the strength of association between categorical variables—in this case, the predicted class distributions of the male and female models. Where relevant, Chi-squared tests of independence are performed to test whether the observed differences in model predictions can be statistically attributed to the gender of the label source. Together, these analyses enable a multi-faceted assessment of whether and how classification outcomes diverge along gender lines, offering insights into the magnitude and consistency of potential gender bias encoded during supervised learning.

6. Experimental Analysis

This section presents the experimental design, evaluation metrics, and the main findings regarding the impact of gender-specific annotations on sentiment classification. First, we describe the process of synthetic dataset creation, in which we simulate varying degrees of annotation consensus between male and female annotators. These datasets serve as a controlled foundation for examining how classifier behavior changes as inter-annotator agreement fluctuates. We then describe the computational environment and the statistical tools employed to assess inter-model agreement and annotation divergence. Finally, we report comparative results for four classification models, analyzing patterns of disagreement between gender-specific training regimes and interpreting the implications for model fairness and bias propagation.

6.1. Synthetic Dataset Construction

To systematically explore the relationship between annotation consensus and model agreement, we created 12 synthetic SA datasets following Algorithm 1 in Section 4. Each dataset is labeled using both male and female annotations and is named following the convention `SynSA-XX`, where `XX` denotes the percentage of agreement between male and female annotators (ranging from 50 to 99). For example, `SynSA-80` corresponds to a dataset where 80% of the labels are identical across genders.

Each dataset contains 999 text instances, uniformly constructed to simulate sentiment classification tasks with short documents ranging from 20 to 100 words. Importantly, each instance is associated with two labels—one from a simulated male annotator and another from a simulated female annotator—allowing for controlled training and evaluation of models on gender-partitioned annotation sources. This approach enables investigation of how disagreement between annotators influences model predictions, independent of confounding factors such as content variability or document length.

6.2. Experimental Settings and Interpretation Guidelines

All experiments were conducted using Google Colab, employing Python version 3.11.13 and the Scikit-learn library version 1.6.1. This consistent and reproducible computational environment ensured standardization across all experimental runs, minimizing potential variability due to software dependencies or hardware inconsistencies.

To evaluate the divergence between male and female model predictions and assess the underlying gender-related annotation biases, we employed several statistical measures. The percentage of consensus quantifies the proportion of identical labels assigned by male and female annotators, offering a first indicator of inter-annotator agreement. Cohen's Kappa (k) measures agreement corrected for chance, following the guidelines of (Landis and Koch 1977): $k \geq 0.81$ indicate almost perfect agreement; 0.61–0.80 substantial; 0.41–0.60 moderate; 0.21–0.40 fair; and below 0.20 slight or poor agreement.

In addition, to explore the statistical relationship between the categorical outputs, we performed a chi-square (χ^2) test of independence. Large χ^2 values indicate significant deviation from the null hypothesis of independence—i.e., a stronger association between classifications. The corresponding p-value quantifies the probability of observing such association under the assumption of no relationship, with $p < 0.05$ typically considered statistically significant.

Finally, Cramér's V was computed to assess the strength of association between male and female model predictions. Its values were interpreted based on thresholds proposed by Alan and Duncan (1997): associations below 0.20 were considered very low, between 0.20–0.39 low, 0.40–0.69 moderate, 0.70–0.89 high, and above 0.90 very high.

7. Results and Analysis

The graph presented in Fig. 2 illustrates the Cohen's Kappa values computed between male and female annotators (annotators curve), as well as the inter-model agreement for each of the four classifiers evaluated—Logistic Regression (LR), Naive Bayes (NB), SVM, and Random Forest (RF)—across synthetic datasets with decreasing levels of annotator concordance. These curves allow for a comparative analysis of how different algorithms respond to gender-related annotation discrepancies.

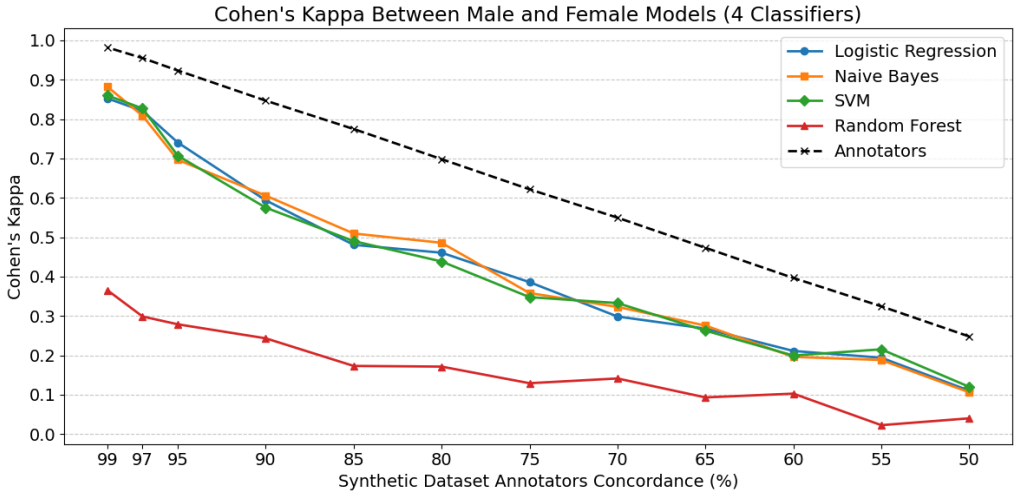

Figura 2. Inter-model agreement (Cohen's Kappa) between classifiers trained on male and female labels across synthetic datasets and synthetic human annotator agreement.

First, it is evident that the Kappa values between models trained with male and female labels fluctuate in accordance with the original concordance of the human annotators. However, none of the classifiers exhibited a behavior fully aligned with the annotator curve. This discrepancy suggests that classifiers internalize and amplify, in distinct ways, the biases embedded in the original annotations. This phenomenon has been previously identified by (Kiritchenko and Mohammad 2018), who demonstrated that SA systems often exhibit gender bias even in simple tasks, attributing different emotional intensities to otherwise identical sentences that differ only in gender references.

Furthermore, the differences observed across models indicate that the architecture of the classifier directly influences how annotated patterns are learned and reproduced. For instance, while NB appears to maintain a more stable variation in the divergence between male and female models, SVM displays more pronounced fluctuations. This aligns with findings by (Kumar et al. 2020), who argue that demographic attributes such as gender and age shape how sentiments are expressed and interpreted by machine learning models. Thus, it becomes crucial to account for the choice of architecture and the feature extraction methods employed.

A complementary statistical analysis of the results provides further insight into how classifiers respond to varying levels of annotation concordance across synthetic datasets, as depicted in Table 1. Beginning with SynSA-99, which exhibits the highest annotator consensus (98.80%), we observe that models trained on male and female labels produce highly concordant outputs, as evidenced by a Cramér's V of 0.982 and a Cohen's Kappa of 0.981. The chi-square statistic, although extremely high (1926.97), confirms that these predictions are significantly dependent, as indicated by the p-value effectively equal to zero. Such strong statistical association suggests a nearly perfect alignment between gender-specific models in scenarios where the data provides little ambiguity—a behavior consistent with high annotator agreement.

As we descend to datasets such as SynSA-90 and SynSA-80, while consensus remains high (89.79% and 79.88%, respectively), Cramér's V declines slightly (0.851 and

Tabela 1. Agreement metrics between male and female labels using NB classifier.

Dataset	Consensus (%)	Cohen's Kappa	Chi²	p-value	Cramér's V
SynSA-99	98.80	0.982	1926.967	0.000	0.982
SynSA-97	97.00	0.955	1823.564	0.000	0.955
SynSA-95	94.89	0.923	1707.568	0.000	0.924
SynSA-90	89.79	0.847	1446.996	0.000	0.851
SynSA-85	84.98	0.775	1227.587	0.000	0.784
SynSA-80	79.88	0.698	1022.289	0.000	0.715
SynSA-75	74.77	0.622	844.656	0.000	0.650
SynSA-70	69.97	0.550	702.965	0.000	0.593
SynSA-65	64.86	0.473	579.080	0.000	0.538
SynSA-60	59.76	0.396	481.834	0.000	0.491
SynSA-55	54.95	0.324	413.877	0.000	0.455
SynSA-50	49.85	0.248	367.889	0.000	0.429

0.715), and chi-square values, although still large and significant, decrease proportionally. This progressive reduction in statistical association signals the early onset of divergence between male and female model predictions. Despite the underlying textual content being constant, minor differences in the gendered labels begin to impact how models interpret and categorize sentiment. This observation is particularly relevant to fairness-aware NLP, as it reveals how even small deviations in annotation standards can initiate bias amplification during training, as highlighted by (Jiang et al. 2024) and (Levshina et al. 2024).

Further along the continuum, in mid-range datasets such as SynSA-70 and SynSA-60, we see a marked shift. Here, Cramér's V values drop to 0.593 and 0.491, indicating only modest levels of agreement between gendered models. The chi-square statistics remain statistically significant, but the reduced strength of association suggests that classifiers are learning and reproducing gender-specific annotation patterns in increasingly divergent ways. Importantly, these levels of model disagreement do not necessarily correspond to low-quality annotations; rather, they may reflect the inherent subjectivity present in SA tasks, which becomes more pronounced in settings with moderate annotator consensus.

In the datasets with the lowest agreement among human annotators—SynSA-55 and SynSA-50—the statistical signals confirm gender-driven model divergence. Cramér's V values fall below 0.46, and although chi-square values (413.88 and 367.89) remain highly significant, the patterns captured by each model are clearly driven by gender-specific annotation differences. Notably, in SynSA-50, where consensus is below 50%, the classifier trained on male labels often disagrees sharply with the one trained on female labels, despite using the same feature representation and algorithm. This confirms that supervised models do not merely reflect ambiguity—they transform it into structured, and potentially biased, prediction behavior. Such findings echo concerns raised by (Kiritchenko and Mohammad 2018) and (Kenyon-Dean et al. 2018) about the limitations of conventional evaluation pipelines in identifying and mitigating representational harms.

As human synthetical annotations become less consistent, classifiers trained on gendered data increasingly diverge, even under identical experimental settings. Thus, consensus, chi-square significance, and Cramér's V collectively offer a triangulated pers-

pective on the fidelity and fairness of gender-based classification behavior. These findings reinforce the call for model auditing practices that move beyond accuracy metrics, incorporating statistical association measures to assess the robustness and ethical implications of sentiment models in real-world applications.

8. Conclusions and Future Work

This study addressed a critical and often underexplored aspect of supervised machine learning in sentiment analysis: the influence of annotator identity—specifically gender—on the behavior of classification models. While large-scale sentiment datasets are routinely constructed using aggregated human labels, little consideration is given to the diversity or representativeness of those annotators. As highlighted in prior works such as (Kiritchenko and Mohammad 2018), annotation bias can significantly shape model behavior, yet standard practices in dataset construction rarely account for this factor explicitly.

Motivated by this gap, we proposed a controlled synthetic dataset generation methodology that simulates sentiment labels from male and female annotators with tunable levels of agreement. We then trained four classification algorithms on gender-partitioned annotations and evaluated their predictions divergence. Our results, measured through inter-model Cohen's Kappa, chi-square tests, p-values, and Cramér's V, reveal that as annotator disagreement increases, so does the divergence in model behavior, suggesting that classifiers do not simply absorb ambiguous labels but structurally encode them into divergent decision boundaries that may amplify latent annotation biases.

These findings resonate with recent concerns in the NLP community regarding the ethical limitations of widely adopted benchmarks. For instance, (Kenyon-Dean et al. 2018) argue that the subjective nature of sentiment should not be treated as a flaw to be minimized, but as a meaningful expression of sociolinguistic variation. Our study reinforces this perspective: the divergences between gendered classifiers were not random noise, but consistently structured and correlated with the degree of annotator disagreement—indicating that the models internalized socially situated perspectives embedded in the labeling process.

Moreover, our statistical analysis showed that these divergences were not only present but significant. The chi-square values increased proportionally to annotator consensus, with p-values nearing zero across all datasets, confirming that the probability of these results being due to chance is negligible. These results are in line with observations by (Jiang et al. 2024), who demonstrated that even when sentiment labels appear superficially aligned, deeper biases often persist in how models interpret gendered content.

Looking ahead, our work suggests multiple promising avenues for future research. First, it is imperative to construct real-world datasets that not only track annotator identity but also actively balance for demographic representativeness. The use of synthetic datasets in this study allowed for precise experimental control, but real-world applications demand that fairness-aware practices be embedded in dataset curation from the outset.

Finally, we advocate for incorporating disagreement-aware auditing protocols into machine learning development cycles. As suggested by (Luitel et al. 2024), model disagreement—far from being a defect—can serve as a diagnostic tool to surface and interrogate ethical blind spots. Rather than striving for consensus as the sole marker of data quality, future research should build real-world datasets with balanced demographics.

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